# RESEARCH STATEMENT

EDWARD VARVAK

## 1. Summary of my results

My research concerns the structure of Lagrangian fibration on symplectic varieties.

**Definition 1.1.** A normal variety $X$ is **symplectic** if it admits a non-degenerate symplectic form $\sigma \in H^0(X^{\mathrm{sm}}, \Omega^2_{X^{\mathrm{sm}}})$ which extends holomorphically onto a resolution of singularities $\pi : Y \to X$. In this setting, a **Lagrangian fibration** is a proper morphism $f : X \to B$ with connected half-dimensional fibers, such that $\sigma|_{X_b} = 0$ for any fiber $X_b$.

We will need to assume that $X$ is compact for most of our results. Additionally, we will want $X$ to be "irreducible" in a sense, which will be made more precise in (*** link below). To that end, the most general class of symplectic varieties we will work with is the primitive symplectic varieties.

**Definition 1.2.** A compact symplectic variety $X$ is **primitive symplectic** if it satisfies $H^1(X, \mathcal{O}_X) = 0$ and $\sigma$ is unique up to scaling, i.e. $H^0(X, \Omega^2_X) = \mathbb{C}\sigma$.

As we will see (*** link below), when $X$ is a primitive symplectic variety admitting a Lagrangian fibration $f : X \to B$, the smooth fibers $X_b$ are Abelian varieties. My main results further classify the structure of the fibers.

**Theorem 1.3.** *Let $X$ be a primitive symplectic variety of dimension $2n$ admitting a Lagrangian fibration $f : X \to B$. If $f$ is not isotrivial, the Mumford–Tate group of the very general fiber $X_b$ is isogenous to $(\mathrm{Sp}_{2h})^{n/h}$.*

Recall that an Abelian variety is Hodge-generic if it admits no extra Hodge tensors. In this case, it is simple and carries no extra symmetries.

**Corollary 1.4.** *The very general fiber $X_b$ is isogenous to a product of Abelian varieties $A_1 \times \cdots \times A_{n/h}$, where each $A_i$ is Hodge-generic and $A_i$ is not isogenous to $A_j$ for $i \neq j$.*

The most well-studied class of primitive symplectic varieties consists of hyper-Kähler manifolds, which should be thought of as the natural generalization of K3 surfaces.

**Definition 1.5.** A compact complex manifold $X$ is **hyper-Kähler** if $X$ is simply connected and its algebra of global holomorphic forms is generated by a non-degenerate symplectic form: $H^0(X, \Omega^{2p}_X) = \mathbb{C}\sigma^p, \quad H^0(X, \Omega^{2p+1}_X) = 0$.

**Theorem 1.6.** *Let $\mathcal{X}$ be a deformation class of hyper-Kähler manifolds such that*

1. *all members $X \in \mathcal{X}$ have second Betti number $b_2(X) \geq 7$,*
2. *there exists some $X \in \mathcal{X}$ admitting a Lagrangian fibration.*

*Choose $X$ to be general among the hyper-Kähler manifolds admitting a Lagrangian fibration, $f : X \to B$. If $X_b$ is a very general fiber of $f$, then it has Mumford–Tate group $\mathbf{MT}(X_b) = \mathrm{GSp}_{2n}$, i.e. $X_b$ is Hodge-generic.*

These results largely came about through understanding the monodromy of the family $f$. To be more precise, let $f : X^\circ \to B^\circ$ be the restriction of $f$ to its regular points, and let $V = R^1 f_* \mathbb{Z}_{X^\circ}$, viewed as a local system.

**Theorem 1.7.** *Suppose that $X$ is primitive symplectic, $f : X \to B$ a Lagrangian fibration, $V_\mathbb{C} = V \otimes_\mathbb{Z} \mathbb{C}$ the underlying $\mathbb{C}$-representation of $\pi_1(B^\circ)$.*

(1) *If $f$ is not isotrivial, then $V_\mathbb{C}$ is irreducible.*
(2) *If $f$ is isotrivial, then its smooth fibers are isogenous to $E^n$, for $E$ an elliptic curve. Furthermore, if $K$ is the endomorphism field of $E$ (possibly $\mathbb{Q}$), then $V_K = U_1 \oplus U_2$ splits into two geometrically irreducible components.*

**Remark 1.8.** There are a few clarifications to be made regarding the results and contributions. For the non-isotrivial case (1), the later result of Theorem 1.3 determines that a finite-index subgroup of the image of $\pi_1(B^\circ)$ lies in $(\mathrm{Sp}_{2h,\mathbb{Q}})^{n/h}$, and acts on $V_\mathbb{C}$ via a direct sum of standard representations. Such a representation is would not be irreducible, if not for the unaccounted finite part of the monodromy transitively permuting the irreducible factors. For the isotrivial case (ii), the result $X_b \sim E^n$ was first proven in (*** KLM), though we produced an alternative proof of this fact.

## 2. Background and motivation

Much of modern research is directed at understanding the deformation classes of hyper-Kähler manifolds. There are currently four known deformation types: the infinite families $K3^{[n]}$ and $\mathrm{Kum}_n$, and the two sporadic types $OG6$ and $OG10$ (loc. cit. O'Grady). The SYZ Conjecture (cite below) predicts that every deformation class $\mathcal{X}$

## 3. Future directions

Department of Mathematics
University of Chicago
5734 S University Ave
Chicago, IL 60637

*Email address*: varvak@uchicago.edu